

**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2018 - 2019)**COMPUTER
BASED TEST**
आसान है!

PRE-MEDICAL : NURTURE COURSE

PHASE : I & II

Test Type : COMMON TEST

Test Pattern : NEET(UG)

TEST DATE : 19 - 08 - 2018

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only.
If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is 3 Hours and Questions Paper Contains 180 Questions. The Max. Marks are 720.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while 1 mark will be deducted for every wrong answer. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of Blue / Black Ball Point Pen only as the correct answer(s) of the question attempted.
8. Use of Pencil is strictly prohibited.

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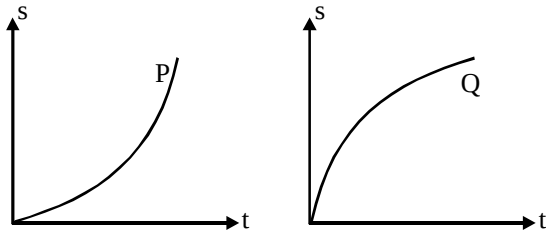
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BEWARE OF NEGATIVE MARKING

1. Two particles P and Q are moving in a straight line along positive x-axis Distance (s) – time (t) graph is plotted for both the particles.



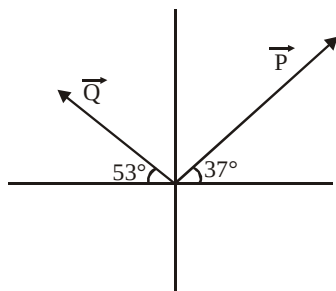
Consider following statements :-

- (a) Speed of P is increasing with time
- (b) Speed of Q is increasing with time
- (c) Average speed of P (during 0 to t) increases with time
- (d) Average speed of Q (during 0 to t) decreases with time
- (e) Acceleration of P is positive.
- (f) Acceleration of Q is negative.

Number of correct statements is :

- (1) 3 (2) 4 (3) 5 (4) 6

2. Find the magnitude of resultant of $\vec{P}$ and $\vec{Q}$ shown in figure. Given $|\vec{P}| = 20$ units and $|\vec{Q}| = 10$ units



- (1) $10\sqrt{5}$ units (2) $10\sqrt{3}$ units
(3) $5\sqrt{10}$ units (4) Zero

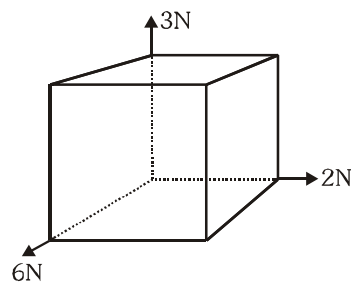
3. A person walks 80 m east, then turns right through angle 143° walks further 50 m and stops. His position relative to the starting point is
- (1) 50 m, 53° east of south
(2) 50 m, 53° south of east
(3) 30 m, 37° south of east
(4) 30 m, 53° south of east

4. An insect starts moving from origin at a constant velocity $(2\hat{i} + 3\hat{j})$ m/s for 1 sec and then with constant velocity $(2\hat{i} + 2\hat{j})$ m/s for $\frac{1}{2}$ sec. The distance of the insect from origin after these displacements is
(1) 4 m (2) 5 m (3) 7 m (4) 10 m

Paragraph for Qus. 5 & 6

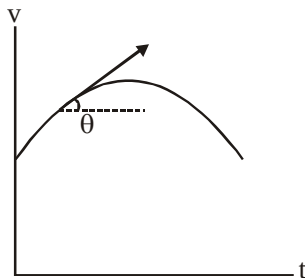
According to Coulomb, any two charges attract or repel each other with a force given as $F = \frac{Kq_1q_2}{r^2}$, where K is called electrostatic constant, q_1 & q_2 are charges & r is the distance between them. Now, we divide a charge Q, into two parts so that when we place them at same distance r, then the electrostatic repulsion between them becomes maximum.

5. The two parts (charges) should be :-
(1) $\frac{Q}{3}, \frac{2Q}{3}$ (2) $\frac{Q}{4}, \frac{3Q}{4}$ (3) $\frac{Q}{2}, \frac{Q}{2}$ (4) None
6. The maximum force will be
(1) $\frac{KQ^2}{r^2}$ (2) $\frac{KQ^2}{4r^2}$ (3) $\frac{KQ^2}{8r^2}$ (4) ∞
7. Three forces of magnitudes 2 N, 3 N and 6 N act at corners of a cube along three sides as shown in figure. Find the resultant of these forces.

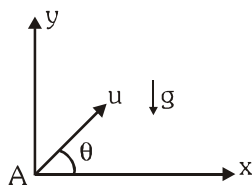


- (1) 11N (2) 9N (3) 7N (4) 5N
8. Two forces $\vec{F}_1$ and $\vec{F}_2$ of equal magnitude produce a resultant of 4 N. When $\vec{F}_2$ is reversed in direction then magnitude of resultant will be of both $2\sqrt{5}$ N. Find magnitude of $\vec{F}_1$.
(1) 2N (2) 3N (3) 4N (4) 5N
9. An object starts from rest at the origin and moves along the x-axis with constant acceleration of 4 m/s^2 . Its average velocity as it goes from $x = 2\text{m}$ to $x = 8\text{m}$ is
(1) 3 m/s (2) 4 m/s (3) 5 m/s (4) 6 m/s

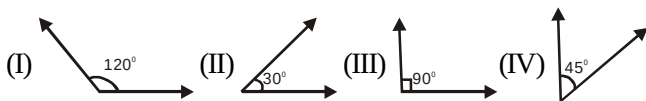
10. Select correct alternative
- (1) Person X moves from location A to location B in 5 seconds. Person Y moves between the same two locations in 10 seconds. Person X is moving with twice the speed as person Y.
 - (2) An object that is accelerating is moving with increasing speed.
 - (3) An object that is moving rightward has rightward acceleration.
 - (4) An object that is moving rightward and speeding up has a rightward acceleration.
11. For a particle moving along x-axis, v-t graph is plotted and a tangent is drawn.



- (1) $\tan \theta$ represent average acceleration.
 - (2) $\tan \theta$ represent instantaneous acceleration.
 - (3) $\tan \theta$ gives only direction of average acceleration, not its magnitude.
 - (4) $\tan \theta$ gives only direction of instantaneous acceleration, not its magnitude.
12. A ball is projected towards right from point A at an angle θ with horizontal. The wind flows towards left (along negative x-direction). It exerts acceleration $g \cot \theta$ on the ball in $-x$ direction. The path of ball will be



- (1) projectile
 - (2) straight line
 - (3) circle
 - (4) none of these
13. Two displacement vectors ($\vec{p}$ and $\vec{Q}$) of same magnitude are arranged in the following manner



Which of the following are correct regarding magnitude of resultant?

- (1) $I = II = III = IV$
- (2) $I > II > III > IV$
- (3) $III > I > II > IV$
- (4) $II > IV > III > I$

Paragraph for Question Nos. 14 to 16

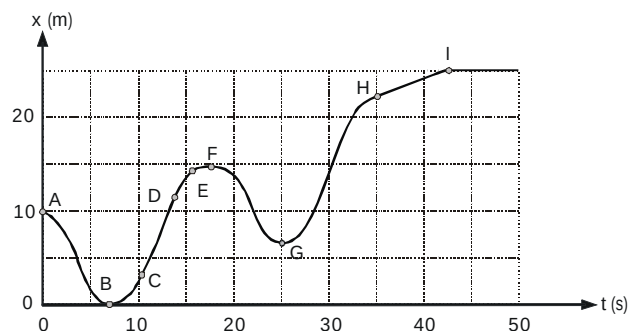
Two boys A and B pass the same place P on a straight road riding their bicycles in the same direction. The boy A passes P at 7:30:00 AM and moves with uniform velocity of 5 m/s. Another boy B passes the place P some time after A, and moves with uniform velocity 10 m/s and overtakes A, 500 m away from P.

14. At what time does the boy B passes place P ?
- (1) 7:30:30 AM
 - (2) 7:30:40 AM
 - (3) 7:30:50 AM
 - (4) 7:31:40 AM
15. Denoting distance traveled in meters by x and time taken in seconds by t , equations of motion of boys A and B respectively are
- (1) $x = 5t$ and $x = 10t + 500$
 - (2) $x = 5t$ and $x = 10t - 500$
 - (3) $x = 5t$ and $x = 10t + 250$
 - (4) $x = 5t$ and $x = 10t - 250$
16. How far ahead of A, would the boy B be at 7:32:30 AM?

- (1) 750 m
- (2) 500 m
- (3) 300 m
- (4) 250 m

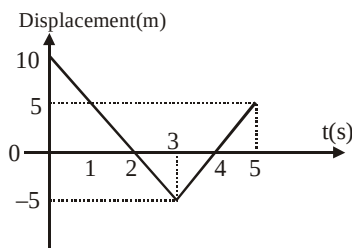
Paragraph for Question 17 to 19

An observer records position of a particle moving on a straight-line path at various instants of time. He starts his stopwatch when the particle is passing the point $x=10$ m. With the help of these data he prepares the following graph, where position x is shown on the ordinate in meters and time t on the abscissa in seconds.

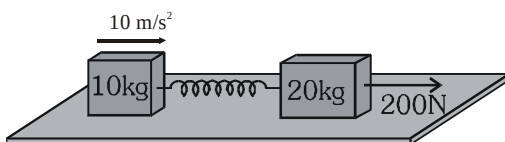


17. At the instant $t = 0$,
- (1) the particle was moving in the negative x-direction and the observer started his stopwatch.
 - (2) the particle was moving in the positive x-direction and the observer started his stopwatch.
 - (3) the particle started its motion from origin with a negative velocity and the observer started his stopwatch.
 - (4) the particle started its motion with a positive velocity and the observer started his stopwatch.

18. Speed of the particle
 (1) first increases then decreases in time interval between points B and F
 (2) first increases then decreases in time interval between points D and F
 (3) always increases between points H and I.
 (4) always decreases between points F and G.
19. The particle is changing its direction of motion at the instant corresponding to points
 (1) A and E only.
 (2) B, F and G only.
 (3) C, D, E and H only.
 (4) C, D, E, H and I only.
20. The diagram shows the displacement–time graph for a particle moving in a straight line. The distance covered by the particle in the time interval $t = 0$ to $t = 5$ s is

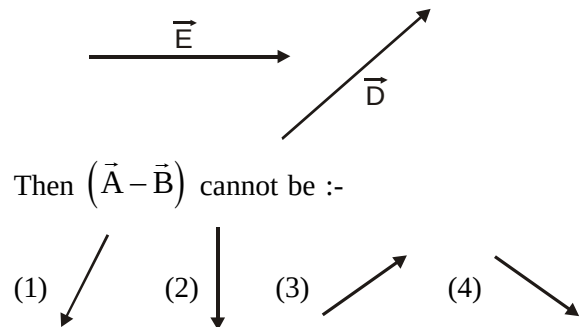


- (1) 12.5 m (2) 17.5 m
 (3) 22.5 m (4) 25.0 m
21. An Indian fighter plane flying horizontally with speed 800 km/hr releases a bomb (on Pakistani bunker) at height of 78.4 m from the ground, when will the bomb strike the ground?
 (1) 2s (2) 4s (3) 6s (4) 8s
22. The masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass. At the instant shown, the 10 kg mass has acceleration 10 m/s^2 then calculate the acceleration of 20 kg mass.



- (1) 5 m/s^2 (2) 10 m/s^2 (3) 15 m/s^2 (4) 20 m/s^2
23. A particle moves along x-axis as $x = u(t - 4) + a(t - 4)^2 + 5$
 (1) The initial velocity of the particle is u .
 (2) The acceleration of particle is a .
 (3) The acceleration of particle is $2a$.
 (4) None of these.

24. A stone falls freely from rest and the total distance covered by it in the last second of its motion equals the distance covered by it in the first five seconds of its motion. The stone remains in the air for :-
 (1) 6 s (2) 13 s (3) 7 s (4) 4 s
25. Two vectors $\vec{A}$ and $\vec{B}$ of unknown magnitudes are along $\vec{E}$ and $\vec{D}$ (as shown below) respectively:



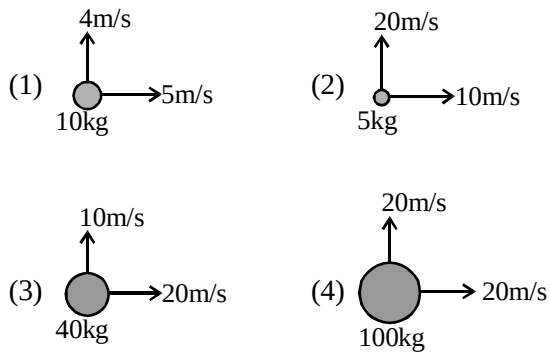
Then $(\vec{A} - \vec{B})$ cannot be :-

- (1) $(2, -6)$ (2) $(8, 0)$
 (3) $(-6, 2)$ (4) $(8, -12)$
26. A butterfly is flying in a garden. First it travels due South for 6m and then 37° West of North for 10m. Assuming that the positive x-direction is East and positive y direction is North, what are the coordinates of the butterfly's final position if butterfly starts from origin :
 (1) $(2, -6)$ (2) $(8, 0)$
 (3) $(-6, 2)$ (4) $(8, -12)$
27. Consider East as +ve x-axis, North as +ve y-axis. A girl walks 10 m East first time then 10m in a direction 30° West of North for the 2nd time & then third time in unknown direction and magnitude, so as to return to her initial position. What is her third displacement vector:-
 (1) $-5\hat{i} - 5\sqrt{3}\hat{j}$ (2) $5\hat{i} - 5\sqrt{3}\hat{j}$
 (3) $-5\hat{i} + 5\sqrt{3}\hat{j}$ (4) she cannot return
28. A particle 'P' starts from point A and goes till point B with velocity $\vec{v}_1$ in time t_1 , then it goes from point B to point C with velocity $\vec{v}_2$ in time t_2 . Another particle 'Q' starts from rest from point A directly towards point C, at the same time as 'P' with a constant acceleration of $\vec{a}$, and takes time t_3 to reach point C. What should be condition(s) in order for P and Q to collide at point C ?
 (1) $\vec{v}_1 t_1 + \vec{v}_2 t_2 = \frac{1}{2} \vec{a} t_3^2$ (2) $t_1 + t_2 = t_3$
 (3) Both (1) & (2) (4) None of the above

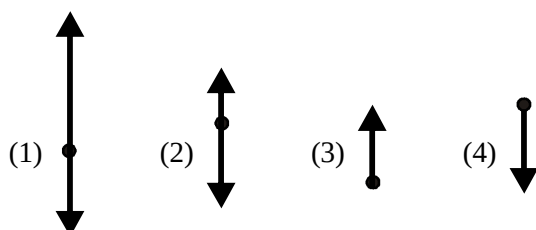
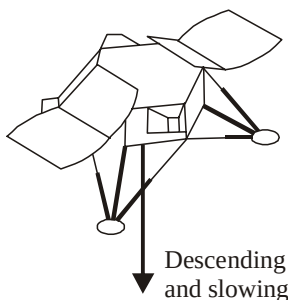
Paragraph for Questions 29 and 30

A ball is rolled on a smooth horizontal surface such that its acceleration $(6\hat{i} - 8\hat{j} + 10\hat{k}) \text{ m/s}^2$. At an instant velocity of ball is $(3\hat{i} + 4\hat{j}) \text{ m/s}$. (x and y-axis lie in horizontal plane in a right handed coordinate system)

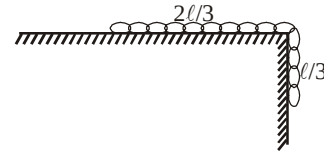
29. Magnitude of component of acceleration along horizontal surface is :-
 (1) 6 m/s^2 (2) 8 m/s^2
 (3) 10 m/s^2 (4) $10\sqrt{2} \text{ m/s}^2$
30. Angle between velocity vector and vertical component of acceleration is :-
 (1) 45° (2) 53°
 (3) 60° (4) 90°
31. Four balls are projected from ground as shown. Which will have maximum range :-



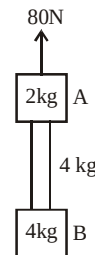
32. A Martian lander is approaching the surface. It is slowing its descent by firing its rocket motor. Which is the correct free-body diagram for the lander? The length of arrow represents the magnitude of the force.



33. An uniform chain is placed over a table such that $\frac{1}{3}$ rd part of it overhangs as shown. If chain remains in equilibrium, then coefficient of static friction between chain and table will satisfy following condition :-

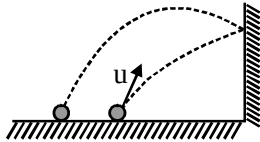


- (1) $\mu \geq 1$ (2) $\mu \geq 0.75$
 (3) $\mu \geq 0.5$ (4) $\mu \geq 0.3$
34. Two objects A and B of masses 2 kg and 4kg are connected by a uniform rope of mass 4kg as shown in the diagram. A force of magnitude 80 N acts on A in vertically upward direction. Tension at mid point of the rope is ($g = 10 \text{ m/s}^2$) :-

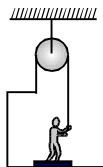


- (1) 32 N (2) 40 N
 (3) 42 N (4) 48 N
35. A ball of mass m is thrown vertically upwards. Air resistance is not negligible. Assume the force of air resistance has magnitude proportional to the velocity, and direction opposite to that of velocity. At the highest point, the ball's acceleration is :
 (1) 0 (2) less than g
 (3) g (4) greater than g
36. Stone is projected from level ground. At a height 0.4 m from the ground, the velocity of the stone in vector form is $\vec{v} = (6\hat{i} + 2\hat{j}) \text{ m/s}$ (the x-axis is horizontal and y-axis is vertically upwards). The angle of projection with horizontal is ($g = 10 \text{ m/s}^2$)
 (1) 45° (2) 60°
 (3) 30° (4) $\tan^{-1}(3/4)$

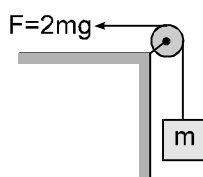
37. A stone is projected from ground and hits a smooth vertical wall after 3 s and again falls back on the ground. The time taken by stone to reach the ground after the collision is 5 s. The maximum height reached by the same stone if the vertical wall were not to be present is. ($g = 10 \text{ m/s}^2$)



- (1) 100 m (2) 80 m
(3) 60 m (4) 50 m
38. A man who can swim at the rate of 2 km/hr (in still river) crosses a river to a point exactly opposite on the other bank by swimming in a direction of 120° to the flow of the water in the river. The velocity of the water current.
- (1) 1 km/hr (2) 2 km/hr
(3) $1/2$ km/hr (4) $3/2$ km/hr
39. Figure shows a man of mass 50 kg standing on a light weighing machine kept in a box of mass 30 kg. The box is hanging from a pulley fixed to the ceiling through a light rope, the other end of which is held by the man himself. If the man manages to keep the box at rest, the weight shown by the machine is .



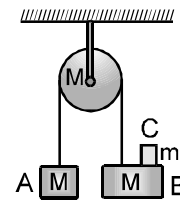
- (1) 500 N (2) 400 N (3) 300 N (4) 100 N
40. The block of mass 'm' equal to 100 kg is being pulled by a horizontal force $F = 2 \text{ mg}$ applied to a string as shown in figure (Take $g = 10 \text{ m/s}^2$). The pulley is massless and is fixed at the edge of an immovable table. What is the value of force exerted by the supporting table on the pulley?



- (1) 707 N (2) 1414 N
(3) 2828 N (4) 3535 N

Paragraph for Que. No. 41-42

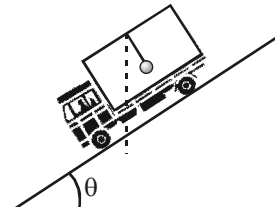
For the following system shown assume that pulley is frictionless, string is massless (m remains on M):



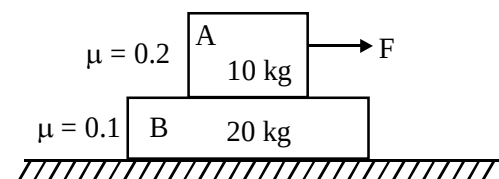
41. The acceleration of the block A is :
- (1) $\frac{mg}{2M+m}$ (2) $\frac{2mg}{2M+m}$
(3) $\frac{mg}{M+2m}$ (4) $\frac{mg}{M+2m}$
42. Normal reaction on m is (force on C due to B).
- (1) $\frac{Mmg}{2M+m}$ (2) $\frac{2Mmg}{2M+m}$
(3) $\frac{Mmg}{M+2m}$ (4) $\frac{2Mmg}{M+2m}$

Paragraph for Que. No. 43-44

A van accelerates uniformly down an inclined hill going from rest to 30 m/s in 6 s. During the acceleration, a toy of mass $m = 0.1 \text{ kg}$ hangs by a light string from the van's ceiling. The acceleration is such that string remains perpendicular to the ceiling. (Take $g = 10 \text{ m/s}^2$)



43. The angle θ of the incline is :
- (1) 30° (2) 60° (3) 90° (4) 45°
44. The tension in the string is
- (1) 1.0 N (2) 0.5 N (3) $\frac{\sqrt{3}}{2} \text{ N}$ (4) $\sqrt{3} \text{ N}$
45. In given diagram what is the minimum value of a horizontal external force F on Block 'A' so that block 'B' will slide on ground is:



- (1) 30 N (2) 20 N
(3) 10 N (4) Not possible

46. Which of the following has the highest mass :
 (1) 1 gram atom of C
 (2) $\frac{1}{2}$ mol of CH_4
 (3) 10 mL of water
 (4) 3×10^{23} atoms of oxygen
47. Number of neutrons in 20 g heavy water are :
 (1) 8 (2) $8 N_A$ (3) 10 (4) $10 N_A$
48. What amount of CaO will be produced by 1g Ca.

$$\text{Ca} + \frac{1}{2} \text{O}_2 \longrightarrow \text{CaO}$$

 (1) 2.8 g (2) 1.4 g (3) 14 g (4) 28 g
49. If 4.9 g H_2SO_4 is present in 100 mL solution. What is the molarity of solution :
 (1) 0.2 M (2) 0.5 M (3) 1 M (4) 0.3 M
50. The angular momentum of electron in an excited H-atom is $\frac{2h}{\pi}$, the K.E. of electron will be:
 (1) +13.6 eV (2) +3.4 eV
 (3) +0.85 eV (4) +0.54 eV
51. d^6 configuration will result in total spin of :
 (1) $\frac{3}{2}$ (2) $\frac{1}{2}$ (3) 2 (4) 1
52. The number of photons of light having wave number x in 10 J of energy source is :
 (1) $10 hcx$ (2) $\frac{ch}{10x}$ (3) $\frac{10}{chx}$ (4) 10
53. The maximum energy is present in an electron at:
 (1) Nucleus
 (2) Ground state
 (3) First excited state
 (4) Infinite distance from the nucleus
54. Isoelectronic species are :
 (1) CN^- , NO^+ (2) CN^- , CO
 (3) CN^+ , NO_2 (4) Both (1) and (2)
55. Which is not in accordance to Aufbau principle:
 (1) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ (2) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$
 2s 2p 2s 2p
 (3) $\boxed{\uparrow}$ $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ (4) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$
 2s 2p 2s 2p
56. One atom of element x weigh 6.64×10^{-23} g. Number of moles of atom in 20 kg x is :-
 (1) 4 (2) 40 (3) 100 (4) 500
57. Which of the following represents isotones ?
 (1) ${}_6\text{C}^{12}$, ${}_6\text{C}^{13}$, ${}_6\text{C}^{14}$ (2) ${}_{18}\text{Ar}^{38}$, ${}_{19}\text{K}^{39}$, ${}_{20}\text{Ca}^{40}$
 (3) ${}_8\text{O}^{16}$, ${}_8\text{O}^{18}$ (4) ${}_{32}\text{Ge}^{77}$, ${}_{33}\text{As}^{77}$
58. How many number of orbitals are possible in L-energy level ?
 (1) 2 (2) 4 (3) 6 (4) 1
59. The radius of second stationary orbit in Bohr's atom is R. The radius of 3rd orbit will be :-
 (1) 9R (2) $\frac{R}{4}$ (3) $\frac{9R}{4}$ (4) 2R
60. If the speed of electron in the Bohr's first orbit of hydrogen atom is x, the speed of the electron in the third Bohr's orbit is :-
 (1) $x/9$ (2) $x/3$ (3) 3x (4) 9x
61. Ψ_{310} has :-
 (1) 1 radial node and 1 angular node
 (2) 2 radial node and 1 angular node
 (3) 1 radial node and 2 angular node
 (4) 2 radial node and 2 angular node
62. Possible number of orientations of a subshell is :-
 (1) ℓ (2) n (3) $2\ell + 1$ (4) n^2
63. What weight of SO_2 can be made by burning sulphur in 5 moles of O_2 gas :-
 (1) 640 g (2) 160 g (3) 80 g (4) 320 g
64. A gaseous mixture contains oxygen and nitrogen in the ratio 1 : 4 by weight therefore, the ratio of number of molecules is :-
 (1) 1 : 4 (2) 1 : 8 (3) 7 : 32 (4) 3 : 6
65. Total gram atoms in 84 g CO are :-
 (1) 3 (2) 6 (3) $3 N_A$ (4) $6 N_A$
66. 11.2 L of N_2 at NTP and 12 g H_2 reacts according to the reaction. Then find out final weight of H_2

$$\text{N}_2 + 3 \text{H}_2 \longrightarrow 2\text{NH}_3$$

 (1) 4.5 g (2) 9 g (3) 18 g (4) 225 g
67. Which of the following statements about an electron with $m_\ell = +2$ is incorrect ?
 (1) The electron may have $m_s = \frac{1}{2}$
 (2) The electron is not in a d orbital
 (3) The electron could be in the third shell
 (4) The electron is in a non spherical orbital
68. Which of the following sets of quantum number is not possible
 (1) $n=3$, $l=+2$, $m_l = 0$, $m_s = +1/2$
 (2) $n=3$, $l=0$, $m_l=0$, $m_s=-1/2$
 (3) $n=3$, $l=0$, $m_l=-1$, $m_s = +1/2$
 (4) $n=3$, $l=1$, $m_l = 0$, $m_s = +1/2$
69. For sodium atom number of electrons with $m = 0$ will be
 (1) 2 (2) 7 (3) 9 (4) 8
70. What is the ratio of radius of 3rd & 5th orbit of Li^{+2} :-
 (1) 3 : 5 (2) 5 : 3 (3) 1 : 1 (4) 9 : 25

71. According to Bohr's theory the approximate quantum number of a circular orbit of diameter 20.6 nm of the hydrogen atom is:-
 (1) 10 (2) 14 (3) 12 (4) 16

72. 12 g of carbon is equivalent to :
 (1) 1 mole C-atom (2) 0.5 mole C-atom
 (3) $\frac{1}{4}$ mole C-atom (4) 2 mole C-atom

73. Number of neutrons present in 8 g atoms of oxygen:
 (1) $64 N_A$ (2) 64 (3) N_A (4) $N_A / 64$

74. An organic compound has the percentage of sulphur as 16%. What is the minimum value of its molecular weight ?
 (1) 50 (2) 200 (3) 8 (4) 32

75. The radial node and angular node for a $4d_{xy}$ orbital will be respectively :-
 (1) 1, 2 (2) 2, 1 (3) 2, 2 (4) 1, 1

76. Calculate the volume of water vapour at STP obtained by the reaction of 4g H_2 with 32g O_2
 $2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)}$
 (1) 4.48 L (2) 2.24 L (3) 22.4 L (4) 44.8 L

77. A metallic oxide contains 60% of the metal. The equivalent weight of the metal is :-
 (1) 12 (2) 24 (3) 48 (4) 72

78. Which of the following statements is **not** correct about the characteristics of cathode rays ?
 (1) They start from the cathode and move towards the anode.
 (2) They travel in straight line in the absence of an external electrical or magnetic field.
 (3) Characteristics of cathode rays do not depend upon the material of electrodes in cathode ray tube.
 (4) Characteristics of cathode rays depend upon the nature of gas present in the cathode ray tube.

79. Which of the following properties of atom could be explained correctly by Thomson model of atom?
 (1) Overall neutrality of atom.
 (2) Spectra of hydrogen atom.
 (3) Position of electrons, protons and neutrons in atom.
 (4) Stability of atom.

80. Which of the following is not an isoelectronic series
 (1) Cl^- , P^{3-} , Ar (2) N^{3-} , Na^+ , Mg^{+2}
 (3) B^{+3} , He, Li^+ (4) N^{3-} , S^{2-} , Cl^-

81. In the first 100 elements number of s-block elements are :-
 (1) 88 (2) 12 (3) 14 (4) 22

82. The electronic configuration of four elements are given below. Which is not true :-
 (i) $[Xe]6s^2$ (ii) $[Xe] 4f^{14}5d^16s^2$
 (iii) $[Ar] 3d^{10}4s^24p^5$ (iv) $[Ar] 3d^74s^2$
 (1) (i) is a strong reducing agent
 (2) (ii) is a d-block element
 (3) (iii) has high negative electron gain enthalpy
 (4) (iv) Show variable oxidation state

83. Which of the following ions has largest size :
 (1) F^- (2) Al^{+3} (3) Cs^+ (4) O^{2-}

84. The process requiring the absorption of energy is:
 (1) $F \rightarrow F^-$ (2) $Cl \rightarrow Cl^-$ (3) $O \rightarrow O^{2-}$ (4) $H \rightarrow H^-$

85. Which of the following diagrams shows correct change in the polarity of bond ?

(1)

(2)

(3)

(4)

86. Select the correct order of acidic strength of oxides.
 (1) $SO_2 > SO_3 > CO_2$ (2) $Cs_2O > K_2O > Na_2O$
 (3) $CO_2 > N_2O_5 > B_2O_3$ (4) $SO_3 > CO_2 > B_2O_3$

87. Choose the incorrect set :
 (I) Bond dissociation energy ($I_2 > Br_2 > Cl_2 > F_2$)
 (II) IE_1 ($Na < Li < B < Be$)
 (III) EA_1 ($N < P < O < S$)
 (IV) Aqueous ion radius ($Na^+ < Mg^{2+} < Al^{3+}$)
 (1) Only I (2) I, II
 (3) I, II, III (4) All of these

88. Identify the correct order of ionisation potential:-
 (1) $Cl > F$ (2) $Al < Ga$ (3) $Li < Na$ (4) $S > P$

89. Correct order of ionic radii is :-
 (1) $Cs^+ > Br^- > S^{2-} > Ca^{+2}$ (2) $Cs^+ > S^{2-} > Br^- > Ca^{+2}$
 (3) $S^{2-} > Br^- > Cs^+ > Ca^{+2}$ (4) $Br^- > Cs^+ > S^{2-} > Ca^{+2}$

90. Element X and Y belong to same group. Which of the following sets of atomic number represents X and Y :-
 (1) 15, 31 (2) 20, 40 (3) 16, 33 (4) 19, 55

- 91.** Find out incorrect matched :
- (1) *Limulus* – King Crab
 - (2) *Locusta* – Gregarious pest
 - (3) *Echinus* – Echinodermata
 - (4) *Taenia solium* – Muscular pharynx
- 92.** Coelom lined up by mesoderm first appeared in the phylum :
- (1) Platyhelminthes (2) Annelida
 - (3) Aschelminthes (4) Arthropoda
- 93.** Intercellular material of cartilage is :
- (1) Solid and pliable (2) Hard and pliable
 - (3) Soft and solid (4) Hard and non-pliable
- 94.** Which of the following is not the function of bone?
- (1) It is the main tissue that provide structural frame to the body
 - (2) To provide support and protection to soft tissue and organs
 - (3) Serve weight bearing functions for body
 - (4) Helps in circulation of different nutrients in different part of body
- 95.** Water transport or canal system is the characteristic feature of :
- (1) Platyhelminthes (2) Coelenterata
 - (3) Echinodermata (4) Porifera
- 96.** Which phylum exhibit two basic body forms called polyp and medusa :
- (1) Porifera (2) Coelenterata
 - (3) Ctenophora (4) Platyhelminthes
- 97.** Aquatic annelida like Nereis possess lateral appendages which helps in swimming are called:
- (1) Setae (2) Parapodia
 - (3) Muscular foot (4) Jointed legs
- 98.** In which phylum mouth contain a file like rasping organ called radula :
- (1) Annelida (2) Arthropoda
 - (3) Mollusca (4) Echinodermata
- 99.** Which of the following is an example of economically important insect ?
- (1) Aedes (2) Anopheles
 - (3) Locust (4) Lacciper
- 100.** Read the following statements carefully which is correct for echinodermata ?
- (1) These animals have an exoskeleton of calcareous ossicles
 - (2) Few are marine with organ system level of organisation
 - (3) The adult echinoderms are radial symmetrical but larvae are bilateral symmetrical
 - (4) They are diploblastic and acoelomate animals
- 101.** Simple cuboidal epithelium with microvilli present in :
- (1) PCT of nephron
 - (2) Loop of Henle's of nephron
 - (3) Collecting duct
 - (4) Bowman's capsule of nephron
- 102.** Germinal epithelium is :
- (1) Simple squamous epithelium
 - (2) Simple columnar epithelium
 - (3) PSCCGE
 - (4) Simple cuboidal epithelium
- 103.** Which junctions help to stop substances from leaking across a tissue ?
- (1) Tight junctions (2) Adhering junctions
 - (3) Gap junctions (4) Interdigitations
- 104.** Ciliated epithelium is present in the inner surface of:
- (1) Blood vessels (2) Oropharynx
 - (3) Fallopian tubes (4) Air sacs of lungs
- 105.** True about compound epithelium is :
- (1) Limited role in secretion and absorption
 - (2) Provide protection against mechanical stresses
 - (3) Cover moist surface of buccal cavity and pharynx
 - (4) All of the above
- 106.** The excess of nutrients which are not used immediately are converted into fats and stored in _____?
- (1) Liver cells (2) Areolar tissue
 - (3) Adipose tissue (4) Lymph nodes
- 107.** Fibroblast, macrophages and mast cells are seen in _____ ?
- (1) Epithelial tissue (2) Areolar tissue
 - (3) Skeleton tissue (4) Neural tissue
- 108.** Cells and fibres are loosely arranged in semifluid ground substance of _____
- (1) Loose connective tissue
 - (2) Dense connective tissue
 - (3) Skeleton connective tissue
 - (4) Blood vascular connective tissue

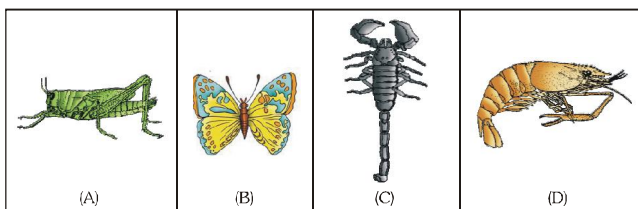
- 109.** Type of tissue found in our heart is :-
 (1) Epithelial and muscular only
 (2) Epithelial and connective only
 (3) Epithelial and nervous only
 (4) All four types of tissues
- 110.** Which statement is not true from the following?
 (1) Dense regular connective tissue is found in periosteum
 (2) Dense irregular connective tissue is found in renal capsule
 (3) Fibres and fibroblasts are compactly packed in dense connective tissue
 (4) Dense irregular is found in wall of blood vessel
- 111.** In the wall of alveoli, which of the following tissue is found ?
 (1) White fibrous connective tissue
 (2) Yellow fibrous connective tissue
 (3) Reticular fibrous connective tissue
 (4) Adipose connective tissue
- 112.** Tendon is an example of [A] tissue and which connect [B] to bone.
 [A] and [B] are respectively :
 (1) Dense regular, Bone
 (2) Dense regular, Muscle
 (3) Dense irregular, Bone
 (4) Dense irregular, Muscle
- 113.** The number of gills present in osteichthyes is :
 (1) 2 pairs (2) 6 pairs
 (3) 5 pairs (4) 4 pairs
- 114.** A jawless fish which lays eggs in fresh water, and whose ammocotes larva after metamorphosis return to the ocean is :
 (1) Labeo (2) Catla
 (3) Myxine (4) Petromyzon
- 115.** Which one of the following is incorrect for aves ?
 (1) Heart is four chambered and animals are oviparous
 (2) The presence of air cavities in bones and the presence of feathers on the body
 (3) Digestive tract has additional chambers and animals are homeothermous
 (4) The forelimbs are not modified into wings

- 116.** Match the following columns :

Column-I		Column-II	
(A)		(i)	Scoliodon
(B)		(ii)	Pristis
(C)		(iii)	Myxine
(D)		(iv)	Catla
		(v)	Petromyzon

- | A | B | C | D |
|--------|-----|-----|-----|
| (1) iv | ii | v | i |
| (2) iv | ii | iii | i |
| (3) i | iii | v | ii |
| (4) i | iv | v | iii |
- 117.** Goblet cells are found in which of the following tissues ?
 (1) Simple cuboidal (2) Stratified squamous
 (3) Glandular epithelium (4) Stratified cuboidal
- 118.** Diffusion boundary of alveoli is formed by ?
 (1) Simple cuboidal
 (2) Simple squamous
 (3) Elasto cartilage
 (4) Transitional epithelium
- 119.** Which of the following tissues are present beneath the skin ?
 (1) Areolar and Tendon (2) Adipose and Tendons
 (3) Areolar and Ligaments (4) Areolar and Adipose
- 120.** Erythropoiesis or RBC formation takes place in ?
 (1) Bone (2) Blood (3) Liver (4) Heart
- 121.** Which of the following is a pair of secretions of exocrine glands ?
 (1) Mucus and hormones (2) Ear wax and hormones
 (3) Saliva and oil (4) Hormones and milk
- 122.** The sequence of organs in chordates from dorsal side to ventral side is :
 (1) Notochord → Nerve cord → Alimentary canal → Heart
 (2) Notochord → Alimentary canal → Nerve cord → Heart
 (3) Nerve cord → Notochord → Alimentary canal → Heart
 (4) Heart → Alimentary canal → Notochord → Nerve cord

- 123.** Well developed larval stage form least developed adult, this type of metamorphosis is shown by :
 (1) Branchiostoma (2) Ascidia
 (3) Petromyzon (4) Climatius
- 124.** Heterocercal tail, Ctenoid scale, 5-7 pair of gills, Operculum, Ampulla of Lorenzini and Air bladder. How many of the following characters are related to elasmobranch fishes ?
 (1) 3 (2) 4 (3) 2 (4) 5
- 125.** Read the given characteristics :
 (a) Megalecithal and Cleidoic eggs, Endothermic
 (b) 12 pairs cranial nerves, metanephric kidney and parental care
 (c) Monocondylic skull and uricotelism
 (d) Cloaca and internal fertilization
 The correct option is :
 (1) All characters are true for aves
 (2) (c) and (d) are true for reptiles, amphibians and platypus
 (3) All characters are true for reptiles and aves only
 (4) All characters are true for aves and platypus
- 126.** Refer the figures A, B, C and D given below. Which of the following option shows the correct name of the animals shown by the figures A, B, C and D ?



- (1) A -Locusta, B ---Butterfly, C ---scorpion, D ---Prawn
 (2) A -Locusta, B ---Prawn, C ---Scorpion, D ---Butterfly
 (3) A -Locusta, B ---Scorpion, C ---Prawn, D ---Butterfly
 (4) A -Butterfly, B ---Scorpion, C ---Prawn, D ---locusta
- 127.** Which of the following statement is not correct related to cyclostomata ?
 (1) Jawless circular mouth
 (2) Body is devoid of scales but paired fins are present
 (3) Cranium and vertebral column are cartilaginous
 (4) Circulation is closed type

- 128.** Which of the following is not a similarity between crocodile and humans?
 (a) four chambered heart
 (b) Thecodont dentition
 (c) corpus callosum
 (d) muscular diaphragm
 (1) a, b (2) a, c (3) b, c (4) c, d
- 129.** Pericardium of heart is composed of
 (1) yellow fibrous connective tissue
 (2) white fibrous connective tissue
 (3) reticular fibrous connective tissue
 (4) areolar connective tissue
- 130.** Which of the following type of tissue is being described by the given statements?
 (a) They are named because of their special functions of linking and supporting.
 (b) They include cartilage, bone, adipose and blood.
 (c) They provide strength and elasticity to the tissues.
 (1) Epithelial tissue (2) Connective tissue
 (3) Muscle tissue (4) Nervous tissue
- 131.** Lack of blood supply and presence of noncellular basement membrane are the characteristic features of the
 (1) muscular tissue (2) fluid connective tissue
 (3) epithelial tissue (4) nervous tissue
- 132.** Which of the following statement is incorrect regarding columnar epithelium?
 (1) Cells are long column like
 (2) Cells consist of basal rounded nucleus
 (3) Found in stomach and intestine
 (4) microvilli may be present on free surface
- 133.** Which one of the following protozoan is considered as most notorious and has staggering effect on human population ?
 (1) Entamoeba (2) Trypanosoma
 (3) Plasmodium (4) Paramoecium
- 134.** All the protozoans are heterotrophs and live as predator or parasite. They are believed to be primitive relatives of animals. There are _____ major groups of protozoans :
 (1) Three (2) Two
 (3) Four (4) Five
- 135.** In which tissue nuclei are located in center and free surface has microvilli :
 (1) Simple columnar (2) Simple cuboidal
 (3) Simple squamous (4) Stratified squamous

136. The lowest category among the seven obligate category is :-

- (1) Genus (2) Species
(3) Family (4) Kingdom

137. Read the following table carefully and select the correct option for P, Q, R and S :

Common name	Biological name	Family	Order
Wheat	<i>Triticum aestivum</i>	Q	R
Mango	P	S	Sapindales

- (1) P-*Oryza sativa*, Q-Poaceae, R-Poales, S-Anacardiaceae
(2) P-*Mangifera indica*, Q-Anacardiaceae, R-Sapindales, S-Poaceae
(3) P-*Oryza sativa*, Q-Sapindales, R-Poaceae, S-Poales
(4) P-*Mangifera indica*, Q-Poaceae, R-Poales, S-Anacardiaceae

138. Reproduction is synonymous with growth is seen in which of the following group of organism ?

- (1) Bacteria, *Amoeba* and Animals
(2) Unicellular algae and All fungi
(3) Bacteria, Unicellular algae and *Amoeba*
(4) Unicellular algae and filamentous algae both

139. Read the statements carefully and select the option which include correct one –

- (I) Herbarium is a store house of collected plant specimen that are dried, pressed and preserved on sheets
(II) Same taxonomic keys are required for each taxonomic category for identification purposes.
(III) Flora provide the index to the plant species found in a particular area
(IV) Monograph contain information on any one taxon.

- (1) I, II and III (2) I, III and IV
(3) I, II, III and IV (4) II, III and IV

140. Which group of plant have embryo but not vascular tissue ?

- (1) Cyanophyta (2) Tracheophyta
(3) Bryophyta (4) Chlorophyta

141. According to three domain theory, how many domain include prokaryotic organism ?

- (1) One
(2) Two
(3) Three
(4) Both (2) and (3) are correct

142. Find the incorrect statement from the following:

- (1) Heterotrophic bacteria are most abundant in nature
(2) Bacterial structure is very complex and they have very simple behaviour
(3) Bacteria reproduce mainly by binary fission
(4) Archaeobacteria differ from other bacteria in having different cell wall structure

143. ____ (A) ____ is a special membranous structure formed by the invaginations of plasma membrane into the cell. These are functionally ____ (B) ____ like structure :

- (1) A-Episome, B-Lysosome
(2) A-Mesosome, B-Mitochondria
(3) A-Sphaerosome, B-Mitochondria
(4) A-Acrosome, B-Lysosome

144. According to R.H. Whittaker's classification system, the group of organism which is heterotrophic eukaryotic, multicellular organism lacking cell wall will be :

- (1) Monera (2) Protista (3) Plantae (4) Animalia

145. Find the incorrect match :

- (1) Thermoacidophiles – Hot sulphur springs
(2) *Mycoplasma* – Smallest cell
(3) Methanogens – Human intestine
(4) *Noctiluca* – Sea ghost

146. The specialized cells of cyanobacteria that fixes atmospheric nitrogen into ammonia is known as :

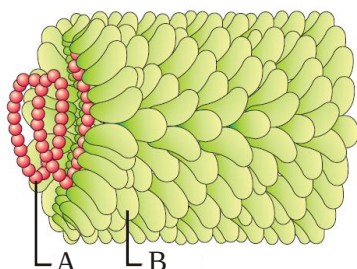
- (1) Auxospore (2) Hormocyst
(3) Heterocyst (4) Elementary bodies

147. Match column-I with column-II and select the correct option :

Column-I		Column-II	
(A)	Chief producer in the ocean	(I)	Euglenoides
(B)	Red tides	(II)	Diatoms
(C)	Mixotrophic nutrition	(III)	Slime mould
(D)	Plasmodium	(IV)	Dinoflagellate

- (1) A-II, B-IV, C-I, D-III (2) A-II, B-IV, C-III, D-I
(3) A-II, B-III, C-I, D-IV (4) A-I, B-IV, C-III, D-II

148. Identify the organism in the given figure and its labelled part A and B :

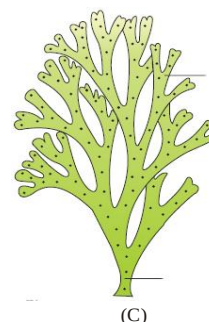
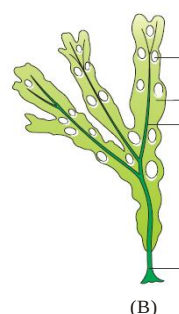
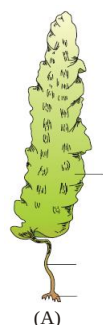


- (1) Bacteriophage, A-DNA, B-Capsid
(2) TMV, A-RNA, B-Tail
(3) Bacteriophage, A-RNA, B-Tail
(4) TMV, A-RNA, B-Capsid
149. Dikaryophase, is shown by :
(1) Phycomycetes and Deuteromycetes
(2) Phycomycetes and Ascomycetes
(3) Basidiomycetes and Deuteromycetes
(4) Ascomycetes and Basidiomycetes
150. In which one of the following the genus name, its character and its class are not correctly matched?

	Genus name	Character	Class
(1)	<i>Trichoderma</i>	Sexual stage is absent	Deuteromycetes
(2)	<i>Aspergillus</i>	Endogenous ascospore	Ascomycetes
(3)	<i>Albugo</i>	Septate and coenocytic mycelium	Ascomycetes
(4)	<i>Phytophthora</i>	Aseptate mycelium	Phycomycetes

151. Which of the following statement is not true ?
(1) Mycelium is generally branched and septate in ascomycetes
(2) Conidia are asexual spores produced endogenously on conidiophores in ascomycetes
(3) Some members of deuteromycetes are saprophytes or parasites while a large number of them are decomposer of litter
(4) Karyogamy and meiosis occur in the basidium in basidiomycetes

152. Consider the following four statements (a–d) and select the option which include all the correct statements only ?
(a) Mycobiont is the major part of lichen thallus
(b) Lichen plays a major role in ecological succession
(c) Lichen act as indicator of SO_2 – pollution
(d) Mycorrhizal association help in N_2 -fixation
(1) Statement (a), (c) and (d) are correct
(2) Statement (b), (c) and (d) are correct
(3) Statement (a), (b) and (c) are correct
(4) Statement (a), (b), (c) and (d) are correct
153. The sex organs of algae are generally :
(1) Unicellular and jacketed
(2) Unicellular and non-jacketed
(3) Multicellular and jacket
(4) Multicellular and non-jacketed
154. Algin and carageenin are obtained respectively from:
(1) Red algae and brown algae
(2) Red algae and green algae
(3) Brown algae and red algae
(4) Brown algae and green algae
155. Pigments present in red algae are :
(1) chl-a, chl-d and fucoxanthin
(2) chl-a and chl-c only
(3) chl-a, chl-d and r-Phycoerythrin
(4) chl-a, chl-c and fucoxanthin
156. Identify the brown algae given below labelled as A, B and C and select the correct option:



- (1) A-Fucus, B-Dictyota, C-Laminaria
(2) A-Dictyota, B-Laminaria, C-Fucus
(3) A-Laminaria, B-Dictyota, C-Fucus
(4) A-Laminaria, B-Fucus, C-Dictyota
157. Life cycle of bryophytes is :
(1) Haplontic life cycle
(2) Diplontic life cycle
(3) Haplo-diplontic life cycle
(4) Diplo-haplontic life cycle

158. The sporophyte made up of foot, seta and capsule in :

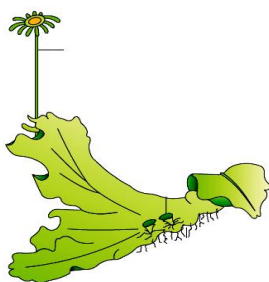
- (1) *Riccia* (2) *Anthoceros*
(3) *Funaria* (4) *Chara*

159. Select the option which include incorrect statements :

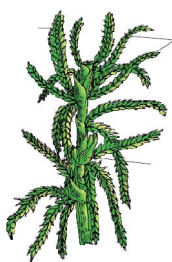
- (I) In *Riccia*, the sporophyte is made up of foot, seta and capsule
(II) The sporophyte of hornworts is completely dependent on its gametophyte
(III) Mosses along with lichens are the first organism to colonise rocks
(IV) *Funaria* possess unicellular unbranched rhizoid

- (1) Only III (2) I, II and III
(3) I, II and IV (4) II and IV

160. Identify the following given figures (A and B) and select true statement about them ?



(A)



(B)

- (1) Figure (A) is *sphagnum* and it is monoecious
(2) Figure (B) is moss and it is dioecious
(3) Figure (B) is *sphagnum* and it is monoecious
(4) Both figure represents moss with dioecious habit

161. Select the incorrect match from the following :-

	Column-I	Column-II
(1)	Golgi complex	Synthesis of glycolipids & glycoproteins
(2)	SER	Steroidal hormones
(3)	Contractile vacuole	Protein synthesis & secretion
(4)	Lysosomes	Nucleic acid digestion

162. How many of the statements are not incorrect ?

- (i) The concentration of ions is significantly lower in the vacuole than in the cytoplasm.
(ii) The cis and trans faces of golgi bodies are entirely different, but interconnected.
(iii) New cells and wanted organs of embryo are destroyed by autophagy of lysosomes.
(iv) Cell sap of vacuoles have fat soluble pigments.

- (1) Two (2) One
(3) Four (4) Three

163. Match the columns and select the correct option:-

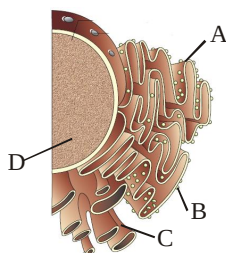
	Column-I		Column-II		Column-III
(i)	Endoplasmic reticulum	a.	Middle men of cell	p.	autolysis
(ii)	Lysosome	b.	System of membranes	q.	Cell sap
(iii)	Golgi apparatus	c.	Tonoplast	r.	Packaging & secretion of materials
(iv)	Vacuole	d.	Suicidal bags	s.	Formation of nuclear membrane

- (1) (i)-b-r; (ii)-d-p; (iii)-a-s; (iv)-c-q
(2) (i)-a-p; (ii)-b-q; (iii)-c-r; (iv)-d-s
(3) (i)-a-s; (ii)-d-q; (iii)-b-r; (iv)-c-p
(4) (i)-b-s; (ii)-d-p; (iii)-a-r; (iv)-c-q

164. Select the incorrect match from the following :-

	Column-I	Column-II
(1)	Vacuole	Storage of water & excretory materials.
(2)	Acid hydrolases	Function in alkaline pH
(3)	Golgi bodies	Polysaccharide synthesis
(4)	RER	Protein synthesis & secretion

165. Select the option with correct labelling of given structure :-



	A	B	C	D
(1)	RER	Nucleus	SER	Ribosomes
(2)	SER	Ribosomes	RER	Nucleus
(3)	RER	Ribosomes	SER	Nucleus
(4)	SER	Nucleus	RER	Ribosomes

166. A number of proteins synthesized by ribosomes on the __A__ are modified in the __B__ of golgi apparatus before they are released from its __C__ face.

- (1) A-Endoplasmic reticulum, B-Cisternae, C-trans
(2) A-Smooth ER, B-Cisternae, C-trans
(3) A-Rough ER, B-Cristae, C-cis
(4) A-Endoplasmic reticulum, B-Cisternae, C-cis

167. Match the columns and select the correct option:-

	Column-I	Column-II	Column-III
(i)	Residual bodies	(a) Formation of acrosome	(p) Detoxification
(ii)	Lysosome	(b) SER	(q) Telolysosomes
(iii)	Golgian vesicle	(c) GERL	(r) Spermiogenesis
(iv)	Cytochrome P ₄₅₀	(d) Exocytosis	(s) ER→Golgi body→lysosomes

- (1) (i)-d-q, (ii)-a-r, (iii)-c-s, (iv)-b-p
(2) (i)-a-p, (ii)-b-q, (iii)-c-r, (iv)-d-s
(3) (i)-c-r, (ii)-d-q, (iii)-b-p, (iv)-a-s
(4) (i)-d-q, (ii)-c-s, (iii)-a-r, (iv)-b-p

168. Synaptonemal complex is observed during cell division in :-

- (1) Mitotic prophase
(2) Meiotic metaphase
(3) Mitotic telophase
(4) Meiotic prophase

169. If there are 26 chromosomes in the nucleus of a frog skin cell, a frog egg would contain ____ chromosomes :-

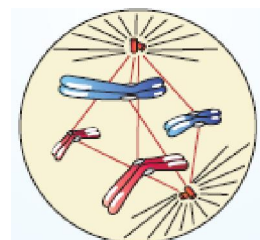
- (1) 13 (2) 26
(3) 39 (4) 52

170. Read the following statements and select how many statements are correct :-

- (A) Crossing over is an enzyme mediated process.
(B) In oocytes of some vertebrates, diplotene can last for months or years.
(C) Variations are not important for the process of evolution
(D) In metaphase I bivalent chromosomes align on the equatorial plate

- (1) Three (2) Four
(3) Two (4) One

171. Identify the given figure and find out correct event of the given phase.



- (1) Anaphase - Chromosome moves towards opposite poles.
(2) Metaphase - Homologous chromosome separates
(3) Transition to metaphase - Chromosome moves towards equator plate
(4) Prophase - Chromatin decondense,

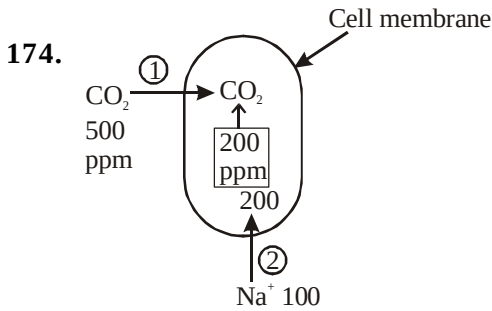
172. Smallest stage of mitosis is ____ in which chromosome number ____ :-

Choose suitable option for above statement ?

- (1) S-phase, remain same
(2) Anaphase, becomes double
(3) S-phase, becomes double
(4) Anaphase, remain same

173. Plant cell wall is made up of :-

- (1) Cellulose, Hemicellulose, Pectins, Proteins
- (2) Cellulose, Galactans, Mannans, CaCO_3
- (3) Only cellulose
- (4) Peptidoglycan



Which type of transport are (1) and (2) ?

- (1) Passive transport and osmosis
- (2) Active transport and passive transport
- (3) Passive transport and active transport
- (4) Passive transport and passive transport

175. Match the following :-

- | | |
|----------------------------|--------------------------------|
| (a) Schleiden | (i) Botanist |
| (b) Schwann | (ii) Zoologist |
| (c) Leeuwenhoek | (iii) Discovery of living cell |
| (d) Robert hooke | (iv) Micrographia |
| (1) a-i, b-ii, c-iii, d-iv | (2) a-i, b-ii, c-iv, d-iii |
| (3) a-ii, b-i, c-iii, d-iv | (4) a-ii, b-i, c-iv, d-iii |

176. Chromatin are made up of :-

- (1) DNA
- (2) DNA, RNA & Histone
- (3) DNA, RNA & Protein
- (4) DNA, RNA, Histone & Non-histone

177. Read following statements :-

- (I) Nucleus is absent in mature RBC & Phloem sieve tube
 - (II) Outer membrane of nucleus is connected with golgi body
 - (III) Euchromatin is genetically less active
 - (IV) Histone made up with acidic amino acid
- Which statement is/are correct ?

- (1) I, II & III
- (2) I, III & IV
- (3) Only I
- (4) Only III

178. Select the correct statement about ribosome :-

- (1) It is a organelle with single membrane
- (2) Composed of rRNA and protein
- (3) 80S ribosome are found in stroma of chloroplast
- (4) Ribosome are present in mammalian RBCs.

179. Lumen of mitochondria is divided into two compartments, outer compartment is called as __A__ and inner compartment is called as __B__.

	A	B
(1)	Matrix	Stroma
(2)	Perimitochondrial space	Matrix
(3)	Matrix	Perimitochondrial space
(4)	Stroma	Matrix

180. Which of the following organelle contains enzyme for β -oxidation?

- (1) Glyoxysome
- (2) Centrosome
- (3) Peroxisome
- (4) Sphaerosome



SPACE FOR ROUGH WORK